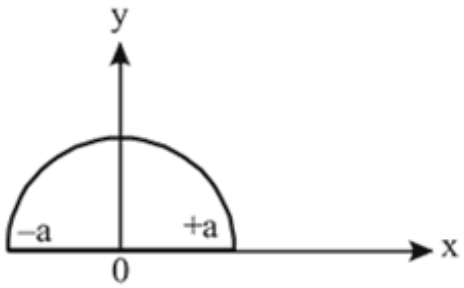
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Determine the rms value of a semi - circular current wave which has a maximum value of a.



(A) $(1\sqrt{2})\mathrm{a}$

(B) $(\sqrt{3/2})\mathrm{a}$

(C) $(\sqrt{2/3})\mathrm{a}$

(D) $(\sqrt{1/3})\mathrm{a}$

Q.2

The r.m.s. value of potential difference V shown in the figure is



(A) V_0

(B) $\mathrm{V}_0/\sqrt{2}$

(C) $\mathrm{V}_0/2$

(D) $\mathrm{V}_0/\sqrt{3}$

Q.3

The equation of alternating current is:
$I = 50\sqrt{2}\sin 400\pi t$ amp. Then the frequency and root mean square value of current are respectively

(A) $200\ \mathrm{Hz}, 50\mathrm{amp}$

(B) $400\pi\mathrm{Hz}, 50\sqrt{2}\mathrm{amp}$

(C) $200\ \mathrm{Hz}, 50\sqrt{2}\mathrm{amp}$

(D) $50\ \mathrm{Hz}, 200\mathrm{amp}$

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Q.4

A coil has resistance 30 ohm and inductive reactance $20\ \text{ohm}$ at $50\ \text{Hz}$ frequency. If an ac source, of 200 volt, $100\ \text{Hz}$, is connected across the coil, the current in the coil will be

(A) $4.0\ \text{A}$

(B) $8.0\ \text{A}$

(C) $\frac{20}{\sqrt{13}}\ \text{A}$

(D) $2.0\ \text{A}$

Q.5

A small signal voltage $V(t) = V_0 \sin \omega t$ is applied across an ideal capacitor C:

(A) Current I (t), lags voltage $\text{V}(\text{t})$ by $90°$.

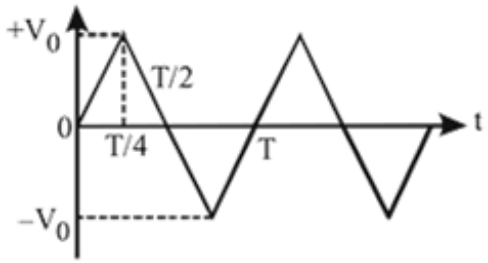
(B) Over a full cycle the capacitor C does not consume any energy from the voltage source.

(C) Current I (t) is in phase with voltage $\text{V}(\text{t})$.

(D) Current I (t) leads voltage $\text{V}(\text{t})$ by $180°$.

Q.6

The voltage time (V-t) graph for triangular wave having peak value V_0 is as shown in figure. The rms value of V in time interval from $\text{t} = 0$ to $\text{T}/4$ is



(A) $\frac{\text{V}_0}{\sqrt{3}}$

(B) $\frac{\text{V}_0}{2}$

(C) $\frac{\text{V}_0}{\sqrt{2}}$

(D) $\frac{\text{V}_0}{3}$

Q.7

An inductor 20mH, a capacitor $50\mu\text{F}$ and resistor 40Ω are connected in series across source of $\text{emf}\ \text{V} = 10 \sin 340\text{t}$. The power loss in A.C. circuit is:

(A) $0.51\ \text{W}$

(B) $0.67\ \text{W}$

(C) $0.76\ \text{W}$

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(D) $0.89\ \mathrm{W}$

Q.8

A resistance 'R' draws power 'P' when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes 'Z', the power drawn will be

(A) $\mathrm{P}\sqrt{\frac{\mathrm{R}}{\mathrm{Z}}}$

(B) $\mathrm{P}\left(\frac{\mathrm{R}}{\mathrm{Z}}\right)$

(C) P

(D) $\mathrm{P}\left(\frac{\mathrm{R}}{\mathrm{Z}}\right)^2$

Q.9

An LCR series circuit with 100Ω resistance is connected to an AC source of $200\ \mathrm{V}$ and angular frequency 300 radians per second. When only the capacitance is removed, the current lags behind the voltage by 60°. When only the inductance is removed, the current leads the voltage by 60°. Then the current and power dissipated in LCR circuit are respectively

(A) $1\ \mathrm{A}, 200$ watt

(B) $1\ \mathrm{A}, 400$ watt

(C) $2\ \mathrm{A}, 200$ watt

(D) $2\ \mathrm{A}, 400$ watt

Q.10

A coil has an inductance of 0.7 henry and is joined in series with a resistance of 220Ω. When the alternating emf of $220\ \mathrm{V}$ at $50\ \mathrm{Hz}$ is applied to it then the phase through which current lags behind the applied emf and the wattless component and the current in the circuit will be respectively

(A) $30^\circ, 1\ \mathrm{A}$

(B) $45^\circ, 0.5\ \mathrm{A}$

(C) $60^\circ, 1.5\ \mathrm{A}$

(D) None of these

Q.11

A resistor of resistance R, capacitor of capacitance C and inductor of inductance L are connected in parallel to AC power source of voltage $\varepsilon_0 \sin \omega t$. The maximum current through the resistance is half of the maximum current through the power source. Then value of R is

(A) $\frac{\sqrt{3}}{\left|\omega C - \frac{1}{\omega L}\right|}$

(B) $\sqrt{3}\left|\frac{1}{\omega C} - \omega L\right|$

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(C) $\sqrt{5} \mid \frac{1}{\omega C}-\omega L \mid$

(D) None of these

Q.12

When the rms voltages $\mathrm{V_L}$, $\mathrm{V_C}$ and $\mathrm{V_R}$ are measured respectively across the inductor L, the capacitor C and the resistor R in a series LCR circuit connected to an AC source, it is found that the ratio is $\mathrm{V_L}: \mathrm{V_C}: \mathrm{V_R}=1: 2: 3$. If the rms voltage of the AC sources is $100\ \mathrm{V}$, the $\mathrm{V_R}$ is close to:

(A) $50\ \mathrm{V}$

(B) $70\ \mathrm{V}$

(C) $90\ \mathrm{V}$

(D) $100\ \mathrm{V}$

Q.13

The current in a series RL circuit decays as $I=I_0 e^{-t/\tau}$. Obtain the rms current in the interval $0 \leq t \leq \tau$.

(A) $\frac{I_0}{2}$

(B) $\frac{I_0}{e}$

(C) $\frac{I_0}{\sqrt{2}\rho}$

(D) $\frac{I_0}{e}\sqrt{\frac{e^2-1}{2}}$

Q.14

An alternating e.m.f. $100 \cos 100t$ volt is connected in series to a resistance of 10 ohm and inductance $100\mathrm{mH}$. What is the phase difference between the current in the circuit and the e.m.f.?

(A) $\frac{\pi}{4}$

(B) zero

(C) π

(D) $\frac{\pi}{2}$

Q.15

An alternating voltage is given by $e=e_1 \sin \omega t+e_2 \cos \omega t$. Then, the root mean square value of voltage is given by

(A) $\sqrt{e_1^2+e_2^2}$

(B) $\sqrt{e_1+e_2}$

(C) $\sqrt{\frac{e_1 e_2}{2}}$

(D) $\sqrt{\frac{e_1^2+e_2^2}{2}}$

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Q.1 Answer:

$(\sqrt{2/3})\mathrm{a}$

Solution:

The equation of a semi - circular wave is

$x^2 + y^2 = a^2$ **or** $y^2 = a^2 - x^2$

$I_{rms} = \sqrt{\frac{1}{2a}\int_{-a}^{+a} y^2 dx}$

$I_{rms}^2 = \frac{1}{2a}\int_{-a}^{+a}\left(a^2 - x^2\right)dx$

$= \frac{1}{2a}\int_{-a}^{+a}\left(a^2 - x^2\right)dx = \frac{1}{2a}\left|a^2x - \frac{x^3}{3}\right|_{-a}^{+a}$

$= \frac{1}{2a}\left(a^3 - \frac{a^3}{3} + a^3 - \frac{a^3}{3}\right) = \frac{2a^2}{3}$

$I_{rms} = \sqrt{\frac{2a^2}{3}} = \sqrt{\frac{2}{3}}a$

Q.2 Answer:

$\mathrm{V}_0/\sqrt{2}$

Solution:

$\mathrm{V}_{\mathrm{rms}} = \sqrt{\frac{(T/2)V_0^2 + 0}{T}} = \frac{V_0}{\sqrt{2}}$

Q.3 Answer:

$200\ \mathrm{Hz}, 50\mathrm{amp}$

Solution:

$2\pi\mathrm{nt} = 400\pi\mathrm{t} \quad \therefore \mathrm{n} = 200 \quad I_0 = 50\sqrt{2}\mathrm{amp}$

r.m.s. current $= I_0/\sqrt{2} = 50\mathrm{amp}.$

Q.4 Answer:

$4.0\ \mathrm{A}$

Solution:

If $\omega = 50 \times 2\pi$ **then** $\omega\mathrm{L} = 20\Omega$

If $\omega' = 100 \times 2\pi$ **then** $\omega'\mathrm{L} = 40\Omega$

Current flowing in the coil is

$$I = \frac{200}{Z} = \frac{200}{\sqrt{R^2 + (\omega' L)^2}} = \frac{200}{\sqrt{(30)^2 + (40)^2}}$$

$$I = 4A$$

Q.5 Answer:

Over a full cycle the capacitor C does not consume any energy from the voltage source.

Solution:

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As we know, power $\mathrm{P}=V_{\mathrm{rms}} \cdot I_{\mathrm{rms}} \cos \phi$
as $\cos \phi=0$
$(\because \quad \phi=90^{\circ})$
$\therefore$ **Power consumed** $=0$ **(in one complete cycle)**

Q.6 Answer:

$\frac{\mathrm{V}_0}{\sqrt{3}}$

Solution:

$V=\frac{V_0}{T/4} t \Rightarrow V=\frac{4V_0}{T} t$
$\Rightarrow \mathrm{V}_{\mathrm{rms}}=\sqrt{<\mathrm{V}^2>}=\frac{4\,\mathrm{V}_0}{\mathrm{T}} \sqrt{<\mathrm{t}\rangle}$
$=\frac{4\,\mathrm{V}_0}{\mathrm{T}}\left\{\frac{\int_0^{\mathrm{T}/4} \mathrm{t}^2 \mathrm{dt}}{\int_0^{\mathrm{T}/4} \mathrm{dt}}\right\}=\frac{\mathrm{V}_0}{\sqrt{3}}$

Q.7 Answer:

$0.51\ \mathrm{W}$

Solution:

Given: $\mathrm{L}=20 \mathrm{mH}; \mathrm{C}=50 \mu \mathrm{F}; \mathrm{R}=40$
$\mathrm{V}=10 \sin 340 \mathrm{t}$
$\therefore \mathrm{V}_{\mathrm{runs}}=\frac{10}{\sqrt{2}}$
$\mathrm{X}_{\mathrm{C}}=\frac{1}{\omega C}=\frac{1}{340 \times 50 \times 10^{-6}}=58.8 \Omega$
$\mathrm{X}_{\mathrm{L}}=\omega \mathrm{L}=340 \times 20 \times 10^{-3}=6.8 \Omega$
Impedance, $\mathrm{Z}=\sqrt{R^2+\left(X_C-X_L\right)^2}$
$=\sqrt{40^2+(58.8-6.8)^2}=\sqrt{4304} \Omega$
Power loss in A.C. circuit,
$\mathrm{P}=i_{\mathrm{rms}}^2 R=\left(\frac{V_{\mathrm{rms}}}{Z}\right)^2 R=\left(\frac{10 / \sqrt{2}}{\sqrt{4304}}\right)^2 \times 40$
$=\frac{50 \times 40}{4304} \simeq 0.51\ \mathrm{W}$

Q.8 Answer:

$\mathrm{P}\left(\frac{\mathrm{R}}{\mathrm{Z}}\right)^2$

Solution:

Power $\mathrm{P}=\mathrm{i}_{\mathrm{rms}}^2 \mathrm{R}=\left(\frac{\mathrm{V}_{\mathrm{rms}}}{\mathrm{R}}\right)^2 \mathrm{R}$
Power when inductor is put in series with $\mathrm{R}: \mathrm{P}'=$

$$\frac{\mathrm{P}'}{\mathrm{P}}=\frac{\left(\frac{\mathrm{V}_{\mathrm{rms}}}{\mathrm{Z}}\right)^2 \mathrm{R}}{\frac{\mathrm{R}}{\mathrm{R}^2} \frac{\mathrm{R}}{\mathrm{R}^2}}=\left(\frac{\mathrm{R}}{\mathrm{Z}}\right)^2$$
$$\mathrm{P}'=\mathrm{P}\left(\frac{\mathrm{R}}{\mathrm{Z}}\right)^2$$

Q.9 Answer:

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2 A, 200 watt

Solution:

Using the foumula,

$$\tan\theta = \frac{\omega L}{R}$$
$$a\tan\theta = \frac{\|\omega c}{R}$$

$$\text{we get, } \omega L = \frac{1}{\omega c}.$$

Impedence of circuit,

$$z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega l}\right)^2} = R$$

current in the circut,

$$I_0 = \frac{V_0}{z} = \frac{V_0}{R} = \frac{200}{100}$$
$$I_0 = 2A$$

Average power, $P = \frac{1}{2} v_0 I_0 \cos\phi$

$$\text{But, } \tan\phi = \frac{\omega L - (1)\omega C)}{R} = 0$$
$$\therefore \quad \cos\phi = 1$$

So,

$$P = \frac{1}{2} \times 200 \times 2 \times 1 = 200W$$

Q.10 Answer:

$45^\circ, 0.5$ A

Solution:

$$\text{L} = 0.7\text{H}, \text{R} = 220\Omega, \varepsilon_0 = 220 \text{ V}, \text{v} = 50 \text{ Hz}$$

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This is an L - R circuit Phase difference,

$$\tan\phi = \frac{\mathrm{X_L}}{\mathrm{R}} = \frac{\omega \mathrm{L}}{\mathrm{R}} = \frac{2\pi \mathrm{vL}}{\mathrm{R}}$$

$$\left[\mathrm{X_L} = 2\pi \mathrm{vL} = 2 \times \frac{22}{7} \times 50 \times 0.7 = 220\Omega\right]$$

$$= \frac{220}{220} = 1 \quad \text{or,} \quad \phi = 45^\circ$$

Wattless component of current

$$= \mathrm{I_0} \sin\phi = \frac{\mathrm{I_0}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \cdot \frac{\mathrm{E_0}}{\mathrm{Z}}$$

$$= \frac{1}{\sqrt{2}} \frac{220}{\sqrt{\mathrm{X_L^2} + \mathrm{R}^2}} = \frac{1}{\sqrt{2}} \cdot \frac{220}{\sqrt{220^2 + 220^2}}$$

$$= \frac{1}{2} = 0.5 \text{ A}$$

Q.11 Answer:

$\frac{\sqrt{3}}{\left|\omega C - \frac{1}{\omega L}\right|}$

Solution:

$\frac{\mathrm{i_{R_0}}}{\sqrt{\left(\mathrm{i_{R_0}}\right)^2 + \left(\mathrm{i_{c_0}} - \mathrm{i_{L_0}}\right)^2}} = \frac{1}{2}$

$\Rightarrow \frac{\epsilon_0/\mathrm{R}}{\sqrt{(\epsilon_0/\mathrm{R})^2 + \left(\epsilon_0 \omega \mathrm{C} - \frac{\epsilon_0}{\omega \mathrm{L}}\right)^2}} = \frac{1}{2}$

$\Rightarrow \mathrm{R} = \frac{\sqrt{3}}{\left(\omega \mathrm{C} - \frac{1}{\omega \mathrm{L}}\right)}$

Q.12 Answer:

90 V

Solution:

Given, $V_L : V_C : V_R = 1 : 2 : 3$

$\mathrm{V} = 100 \text{ V } \mathrm{V_R} = ?$

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As we know, $V = \sqrt{V_R^2 + (V_L - V_C)^2}$

Solving we get, $V_R \simeq 90$ V

Q.13 Answer:

$\frac{I_0}{e}\sqrt{\frac{e^2-1}{2}}$

Solution:

$$i_{rms}^2 = \frac{1}{\tau}\int_0' I^2 dt = \frac{1}{\tau}\int_0^{\tau} I_0^2 e^{-2t/\tau} dt = \left.\frac{-I_0^2 e^{-2t/\tau}}{\tau\frac{2}{\tau}}\right|_0 = \frac{I_0}{e}\sqrt{\frac{e^2-1}{2}} \Rightarrow \quad I_{rms} = \frac{I_0}{e}\sqrt{\frac{e^2-1}{2}}$$

Q.14 Answer:

$\frac{\pi}{4}$

Solution:

$$\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{X_L^2+R^2}}, \quad R = 10\Omega, X_L = \omega L = 10\Omega$$
$$\therefore \quad \cos\phi = \frac{1}{\sqrt{2}} \quad \textbf{or} \quad \phi = \frac{\pi}{4}$$

Q.15 Answer:

$\sqrt{\frac{e_1^2+e_2^2}{2}}$

Solution:

$$e = e_0\sin(\omega t + \phi), \textbf{ where } e_0 = \sqrt{e_1^2+e_2^2}; \quad e_{\text{rms}} = \frac{e_0}{\sqrt{2}} = \sqrt{\frac{e_1^2+e_2^2}{2}}$$

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